

Shocks and adaptation: network resilience, systemic risks

Economic Complexity, Class 9 – Nov 24 2025

Social Data Science, Corvinus University of Budapest

Panel 3: shocks, diffusion, recent EC

8	Nov 18	Presentation of students' research proposal		
9	Nov 25	Gergő Tóth and Balázs Lengyel Shocks and adaptation: network resilience, systemic risks	III. Recent research	[13,1
10	Dec 2	Balázs Lengyel Spatial diffusion of innovation: agents of change, interregional networks, and spatial life cycles		[15,1
11	Dec 9	Viktor Stojkoski Digital trade and Multidimensional Economic Complexity		[17,1
12	Dec 16	Written exam / Presentation of Projects		

[13] Tóth, G., Elekes, Z., Whittle, A., Lee, C., & Kogler, D. F. (2022). Technology network structure conditions the economic resilience of regions. *Economic Geography*, 98(4), 355-378.

[14] Diem, C., Borsos, A., Reisch, T., Kertész, J., & Thurner, S. (2022). Quantifying firm-level economic systemic risk from nation-wide supply networks. *Scientific Reports*, 12(1), 7719.

[15] Elekes, Z., Boschma, R., & Lengyel, B. (2019). Foreign-owned firms as agents of structural change in regions. *Regional Studies*, 53(11), 1603-1613.

[16] Lengyel, B., Bokányi, E., Di Clemente, R., Kertész, J., & González, M. C. (2020). The role of geography in the complex diffusion of innovations. *Scientific Reports*, 10(1), 15065.

[17] Stojkoski, V., Koch, P., Coll, E., & Hidalgo, C. A. (2024). Estimating digital product trade through corporate revenue data. *Nature Communications*, 15(1), 5262.

[18] Stojkoski, V., Koch, P., & Hidalgo, C. A. (2023). Multidimensional economic complexity and inclusive green growth. *Communications earth & environment*, 4(1), 130.

Today

1. Shocks and resilience – EC, Gergő
2. Diffusion – App. NetSci
3. Shock propagation – App. NetSci

DOES RENEWABLE ENERGY TECHNOLOGICAL INNOVATION? CONTROL CHINA'S AIR POLLUTION? A SPATIAL ANALYSIS

Figure 1.

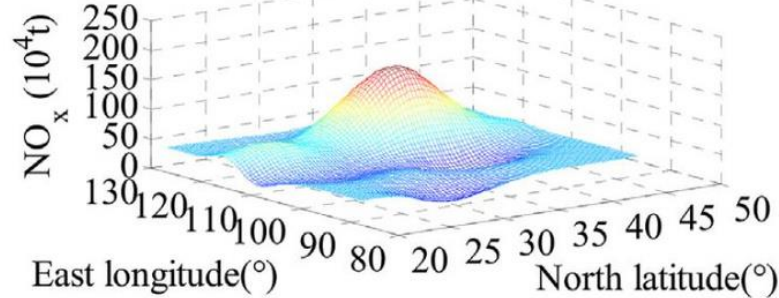
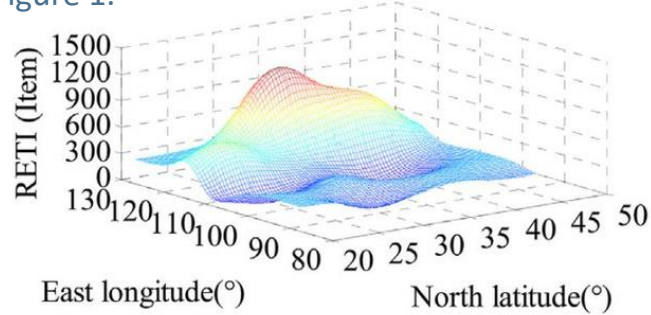


Figure 1. Spatial patterns of RETI and Air Pollution

$$\ln Y_{jt} = \lambda_0 + \lambda_1 \ln RETI_{it} + \lambda_2 \ln POP_{it} + \lambda_3 \ln GDP_{it} + \rho \sum_{j=1}^n W_{ij} \ln Y_{jt}$$

$$\delta_1 \sum_{j=1}^n W_{ij} RETI_{jt} + \delta_2 \sum_{j=1}^n W_{ij} POP_{jt} + \delta_3 \sum_{j=1}^n W_{ij} GDP_{jt} + u_i + v_t + e_{it}$$

Equation 1. Spatial Durbin Model Equation

Results:

- Air Pollution is spatially autocorrelated
- RETI has spatial spillover effects on Air Pollution
- RETI is negatively associated with Air Pollution

Figure 2.

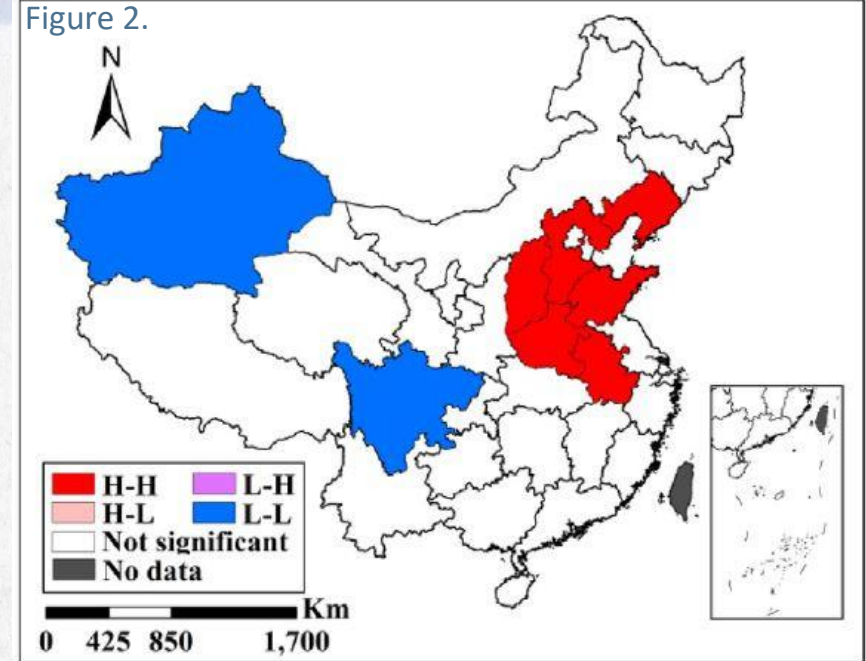


Figure 2. Spatial autocorrelations of RETI

Policy recommendations:

- Support of RETI in the North China Plain
- Establishment of a specialized community to share R&D talent, funds, and resources
- Promotion of industrial upgrade and the adoption of renewable energy sources

Optimized data center site selection – mesoclimatic effects on data center energy consumption and costs.

Energy Efficiency **14**, 33 (2021).

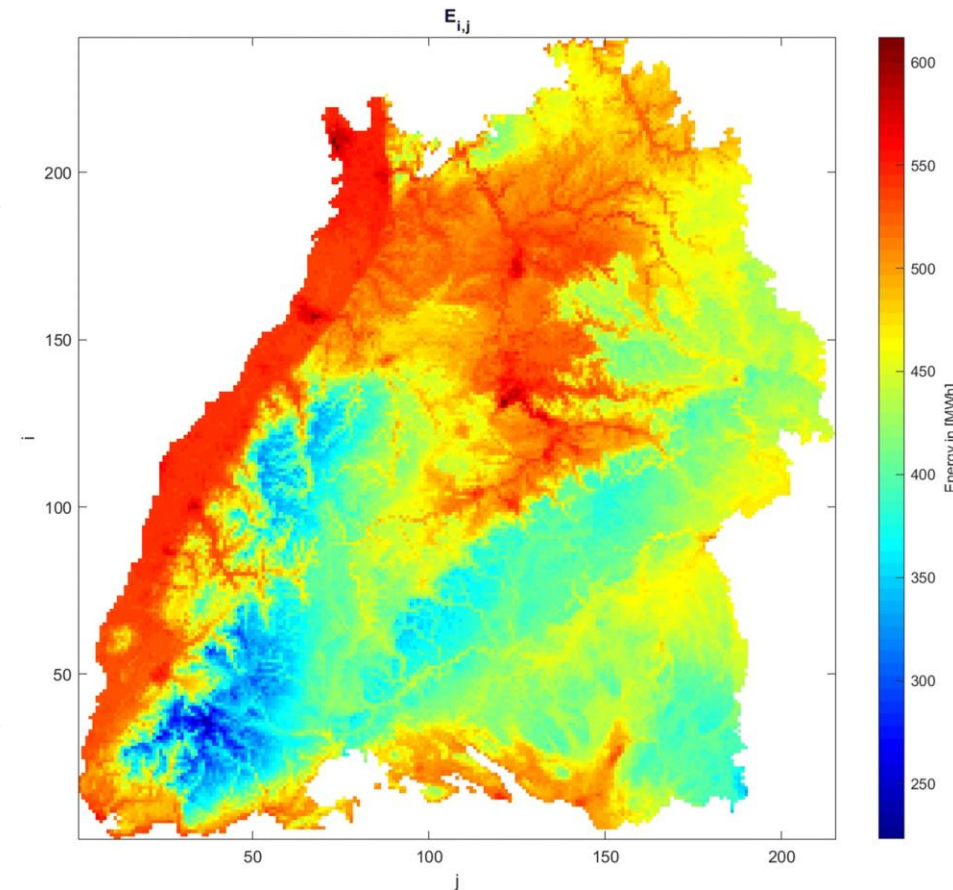
<https://doi.org/10.1007/s12053-021-09947-y>

This paper explores, how climate influences a data center's energy demand, thus revealing one key factor that influences where data centers will cluster regionally. Climate matters, due to the cooling need of data centers, where they can receive 'Free cooling' from the environment.

Data: The paper used energy consumption and heat data of data centers and aggregated temperature data from the German state of Baden-Württemberg.

Method: They created models for calculating energy consumption by location and models of how much energy can data centers save by relocating 10, 25, 100 kms.

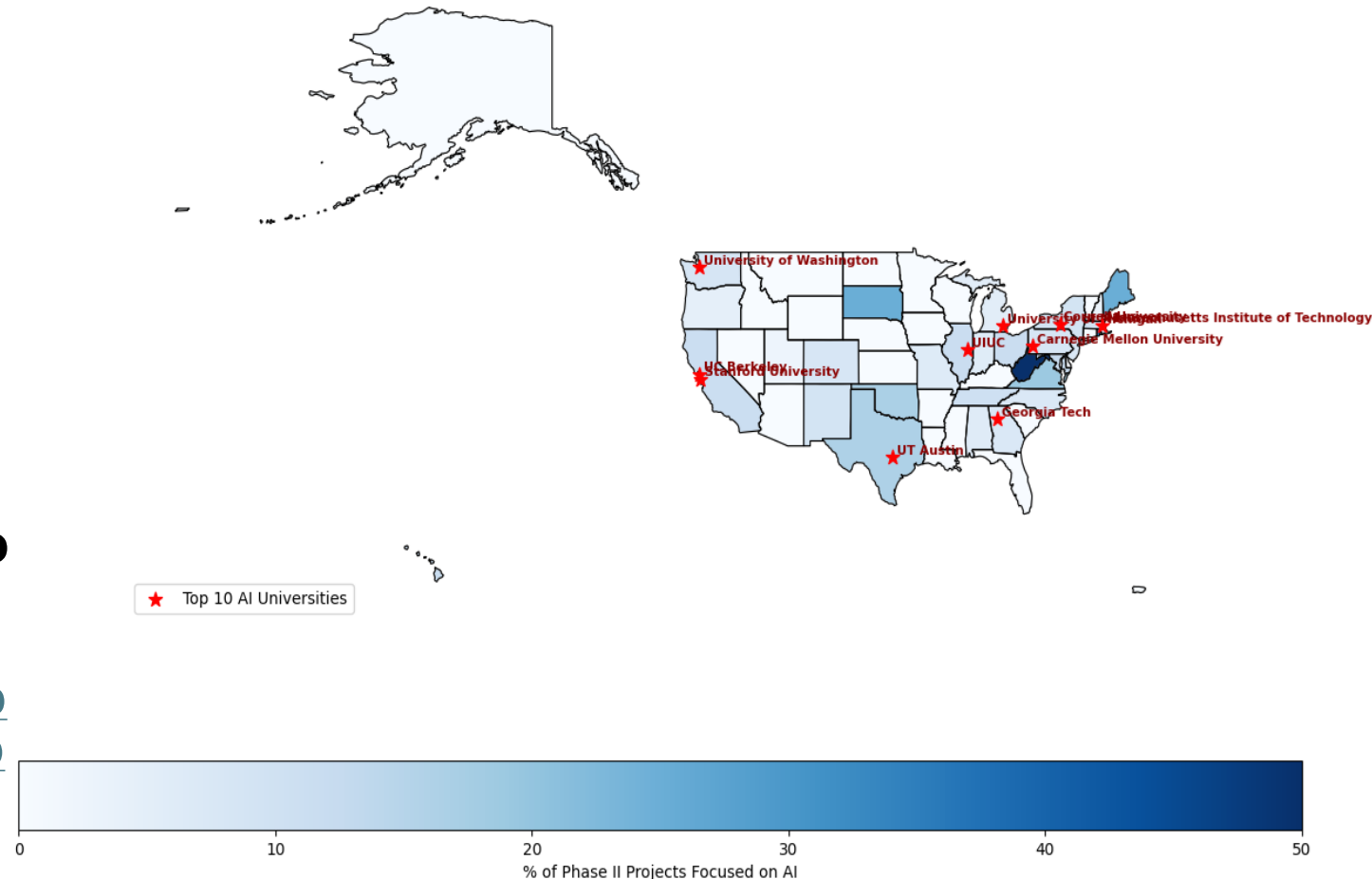
Findings: With on average reducing energy consumption 9.12% when relocating in a 10 km range, and 37.35% relocating in 100 km. Climate plays a crucial role in the decision of a data center's location.



The spatial distribution of public support for AI research By: Farhat Chowdhury, Albert N. Link, Martijn van Hasselt
Chowdhury, F., Link, A. N., & van Hasselt, M. (2022). The spatial distribution of public support for ai research.
Science and Public Policy, 49(4), 573–579. <https://doi.org/10.1093/scipol/scac008>

- **Between US firm awarded for AI research**
- **Top 10 universities ranked by their AI research importance**
- **High correlation in states**
- **Access to AI research resources are fundamental**
- **AI research fundings are geographically concentrated**
- **Policy makers should consider to wider the spatial distribution of funds**
- <https://colab.research.google.com/drive/1kwOFSiDr8MQoEqF48oBhB9gzDSLILSPK?usp=sharing>

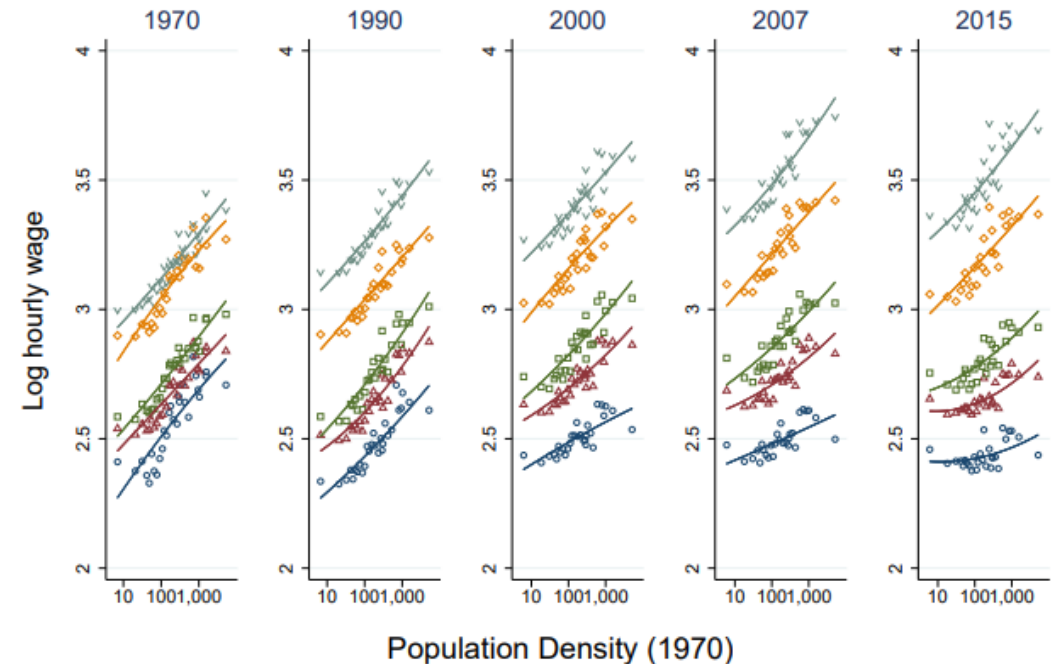
**Spatial Distribution of AI-Focused SBIR Phase II Projects (FY 2020)
with Top 10 AI Universities**



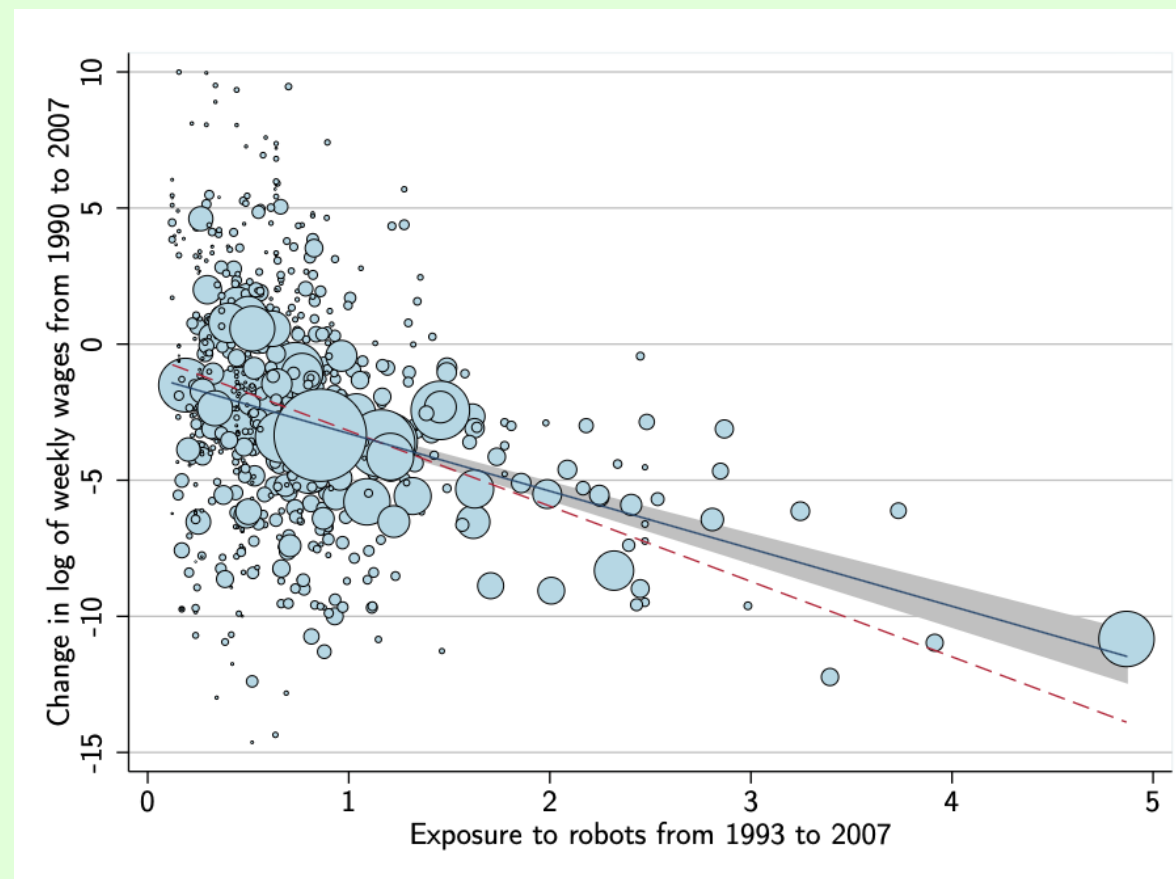
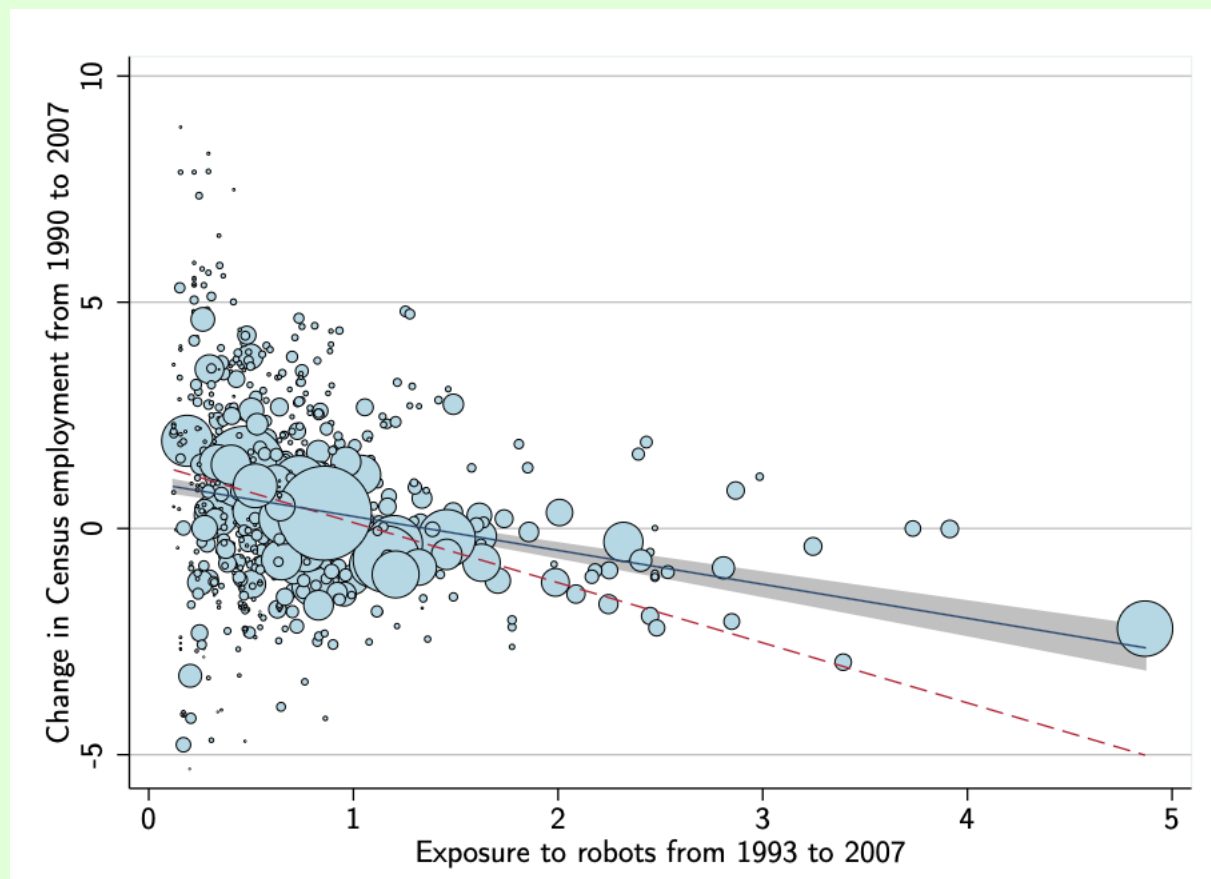
Work of the Past, Work of the Future

by David H. Autor (2019)

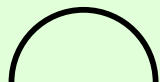
- **Research question:** How has technological change, especially automation, transformed the structure and geography of work in the United States?
- **Data:**
 - U.S. Census and American Community Survey (ACS) microdata (1980-2015)
 - Bureau of Labor Statistics (BLS) occupational employment data
 - O*NET task measures → classify jobs by routine, manual, or abstract task intensity.
 - Covers men and women aged 16-64 across U.S. metropolitan areas.
- **Methodology:**
 - Task-based decomposition: Uses O*NET data to classify occupations by routine vs. non-routine tasks and track employment and wage changes (1980-2015) using U.S. Census and ACS data.
 - Spatial comparison: Examines how automation-driven job polarization differs across city sizes.
- **Results:**
 - Middle-skill routine jobs declined sharply, while high- and low-skill jobs expanded → strong job polarization.
 - Large urban areas captured most high-wage, high-skill growth, leaving smaller regions behind.
 - Automation and digitalization are key forces driving this new geography of work and rising inequality.



Employment and wages vs exposure to robots

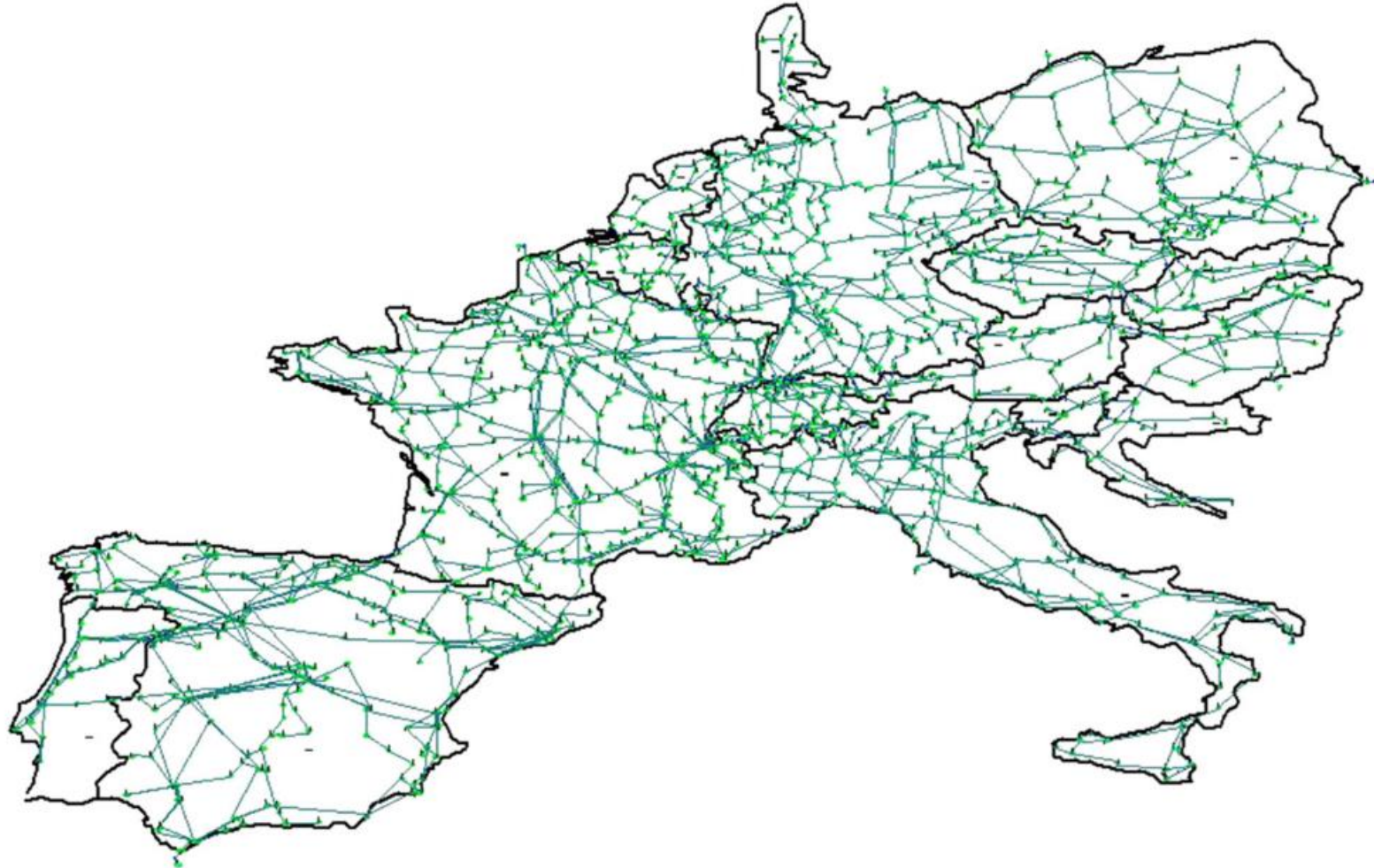


source: *Robots and Jobs: Evidence from US Labor Markets* by Acemoglu and Restrepo



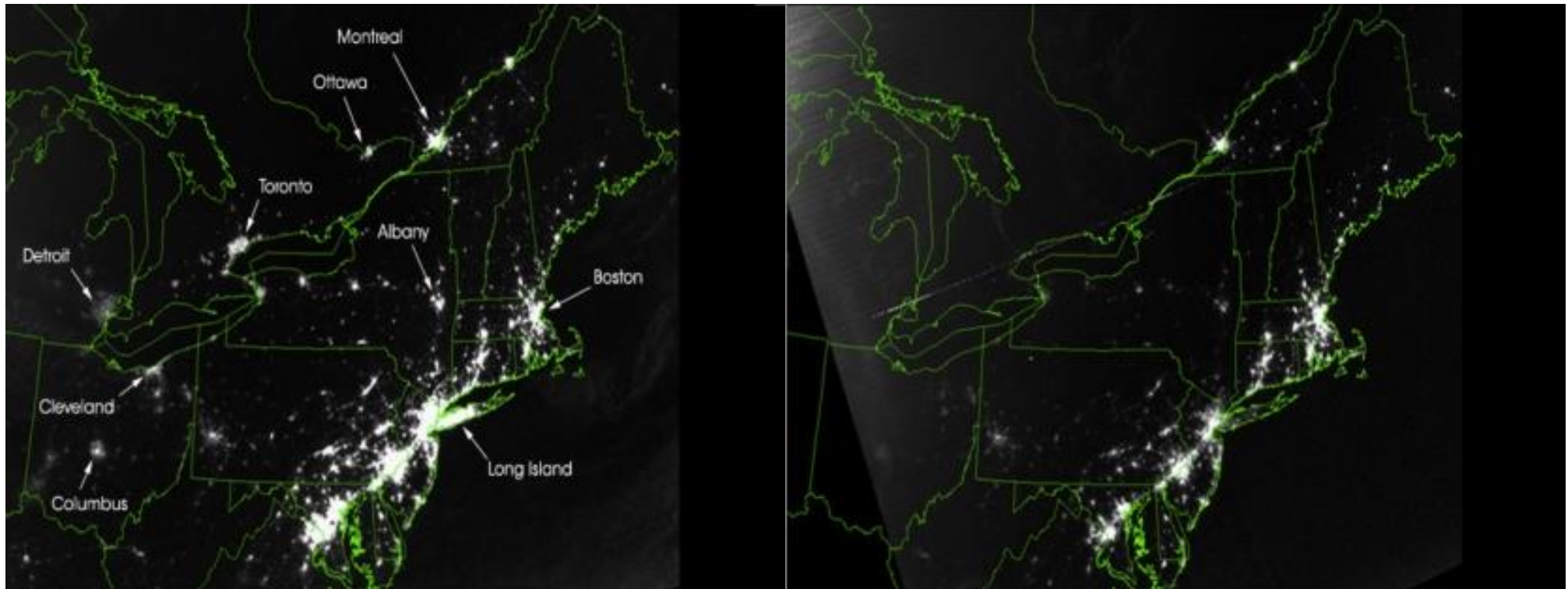
Network resilience and robustness

Gergo Toth
25th November 2025



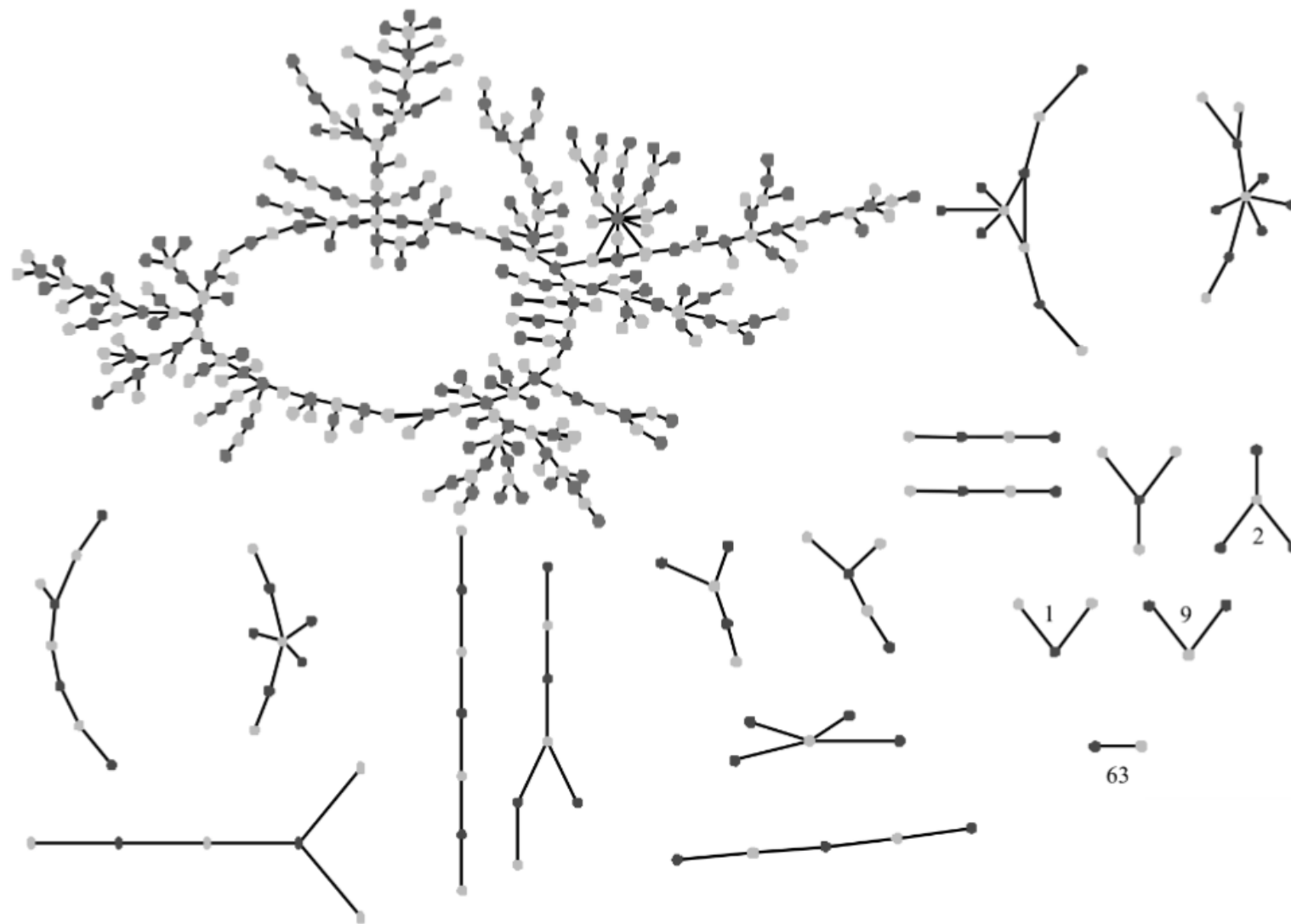
Simplified continental Europe power grid.
(Bompard, E., Pons, E., & Wu, D. 2013).

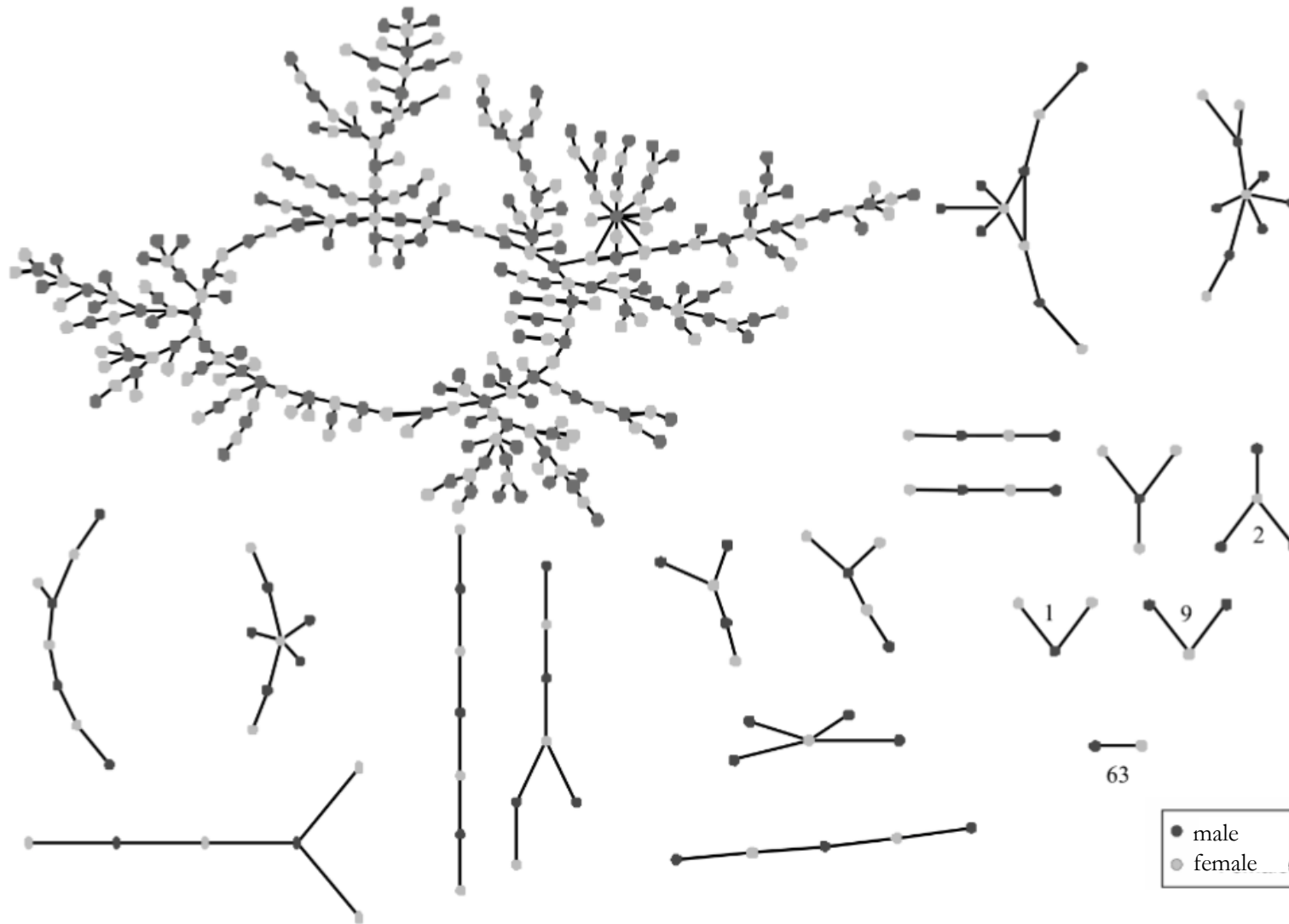
2003 North American Blackout



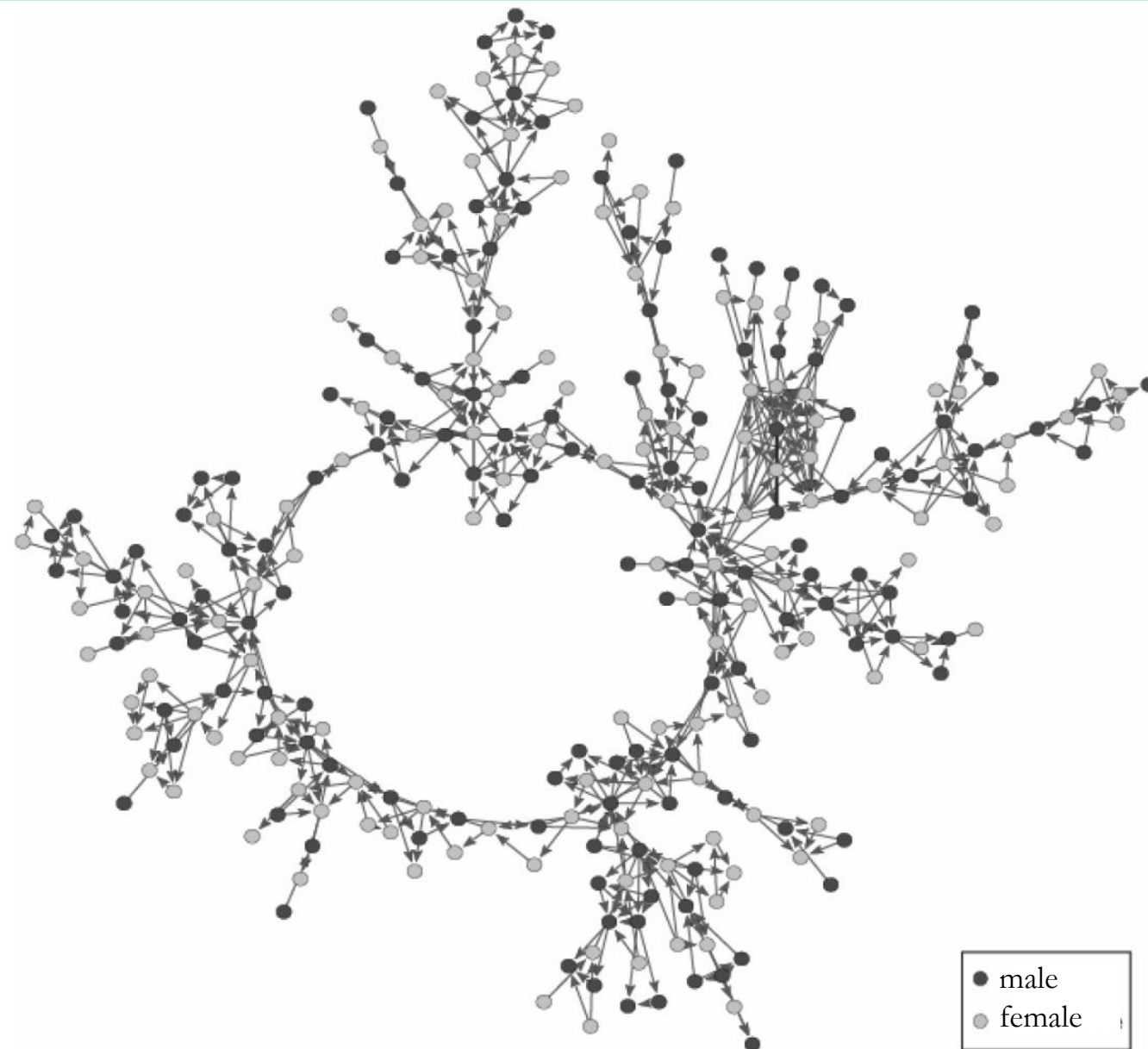
Satellite image on Northeast United States on August 13th, 2003, at 9:29pm (EDT), 20 hours before the 2003 blackout.

The same as before, but 5 hours after the blackout.

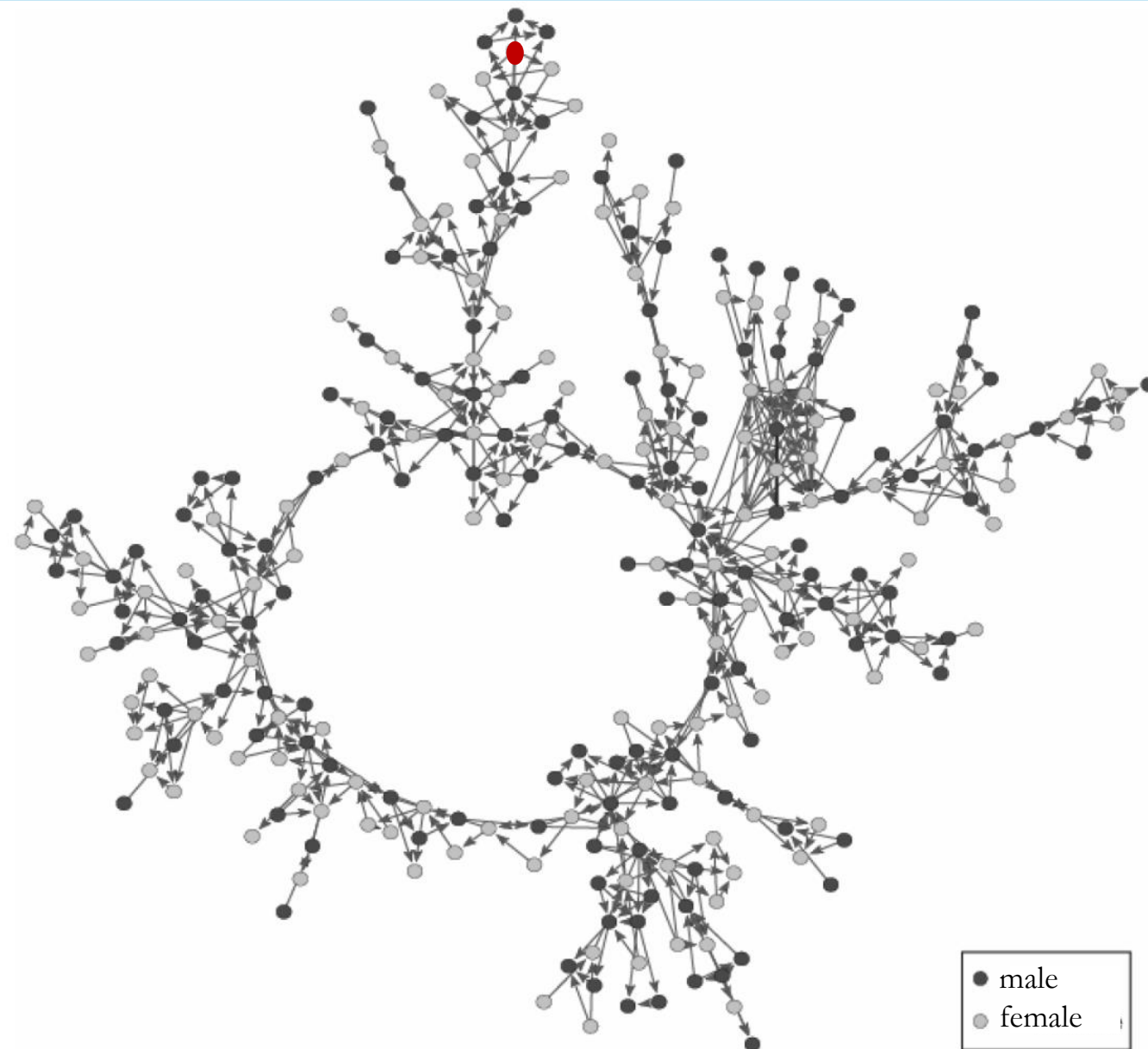




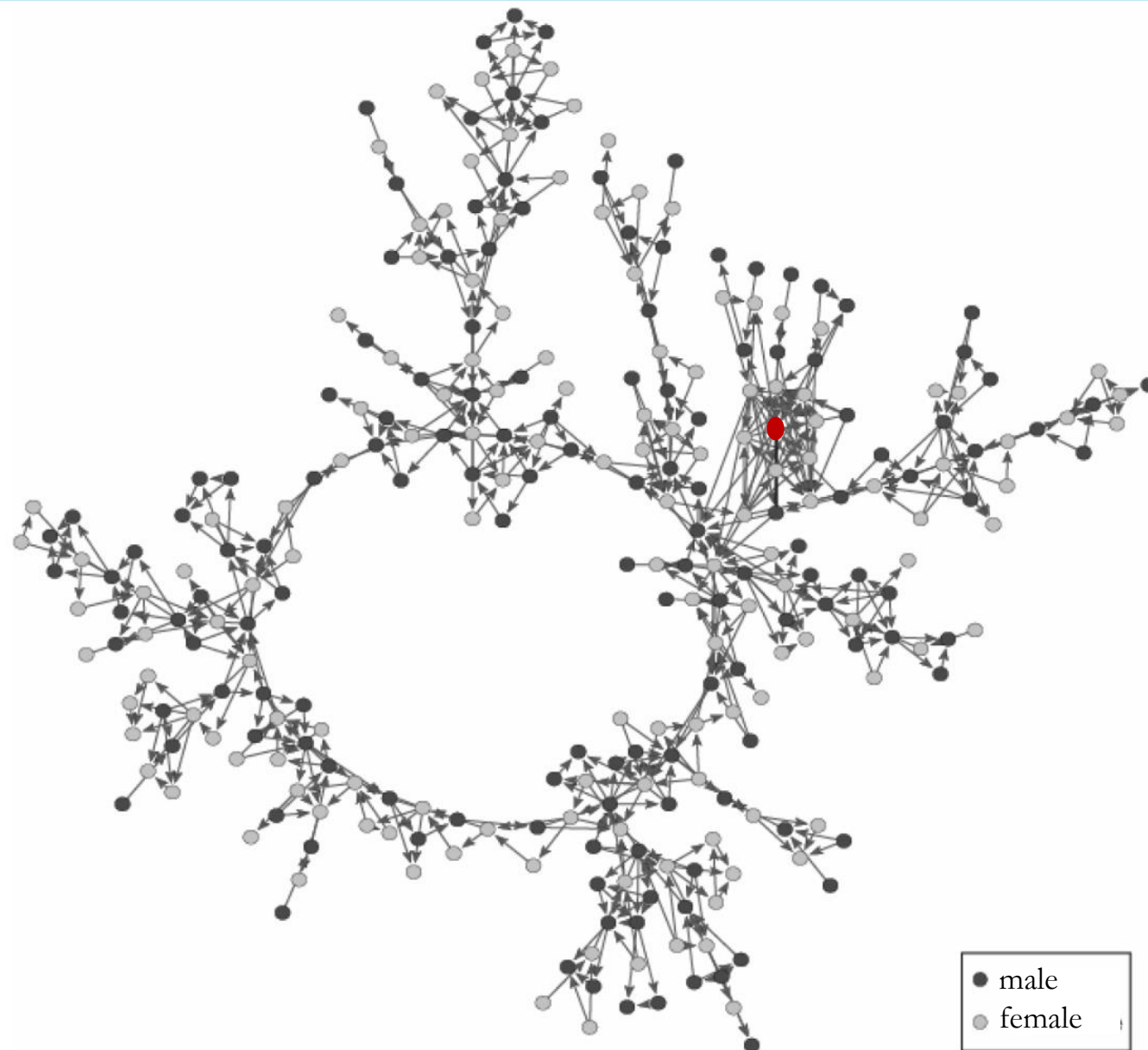
The direct relationship structure at Jefferson High
(Bearman, P. S., Moody, J., & Stovel, K. 2004 AJS)



Simulated minimal temporal connectivity in the Jefferson High partnership network.
(Bearman, P. S., Moody, J., & Stovel, K. 2004 AJS)



Simulated minimal temporal connectivity in the Jefferson High partnership network.
(Bearman, P. S., Moody, J., & Stovel, K. 2004 AJS)



Simulated minimal temporal connectivity in the Jefferson High partnership network.
(Bearman, P. S., Moody, J., & Stovel, K. 2004 AJS)

Why study network resilience/robustness?

Why study network resilience/robustness?

Networks play a key role in the robustness of biological, social and technological systems.

The meaning of clustering, path length, information content of node ranks, communities, etc. all depend crucially on some gestalt notion of connectivity of a network.

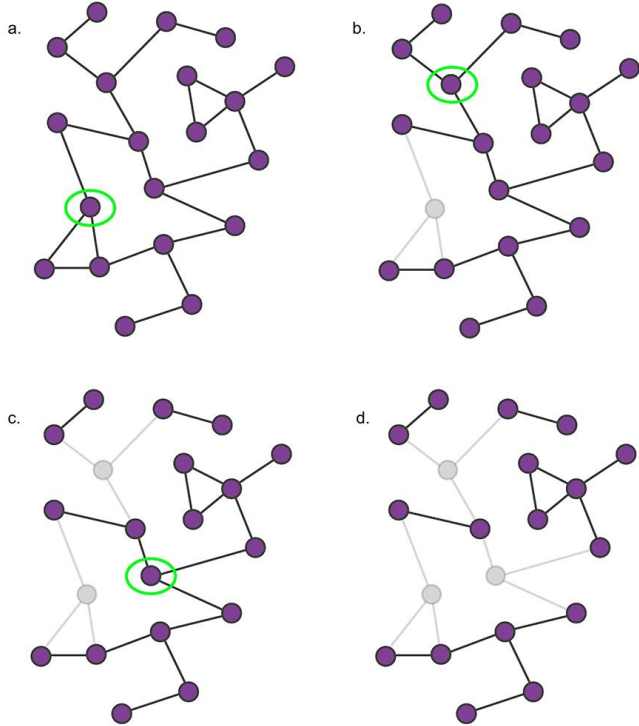
Robustness: a system is robust if it can maintain its basic functions in the presence of internal and external errors.

In a network context robustness refers to the system's ability to carry out its basic functions even when some of its nodes and links may be missing.



Can we measure network resilience/robustness?

When is a network a network?



Barabasi (2016)

Key question: how many nodes do we have to remove to fragment a network into isolated components?

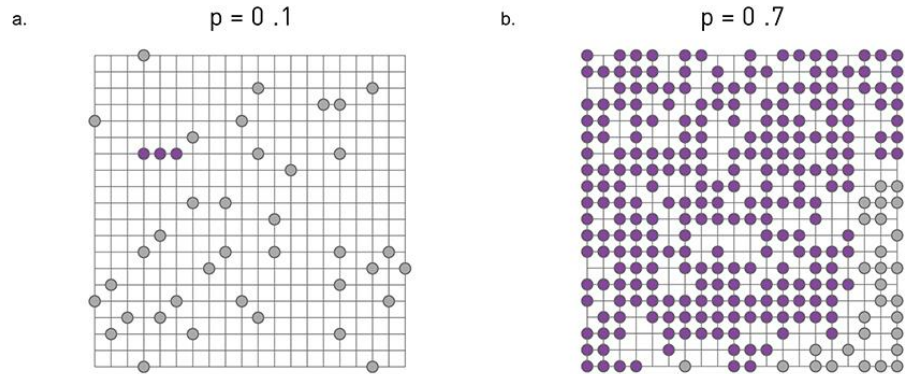
After removing a few nodes, any pair of nodes is still likely to be connected through at least one path.

What could be the conditions for the above statement to be true?

Why does it matter if nodes are connected through at least a path?

Can we measure network resilience/robustness?

When is a network a network?



Percolation theory

a) We place pebbles with probability p at each intersection. Neighbouring pebbles are considered connected, forming clusters of size two or more.

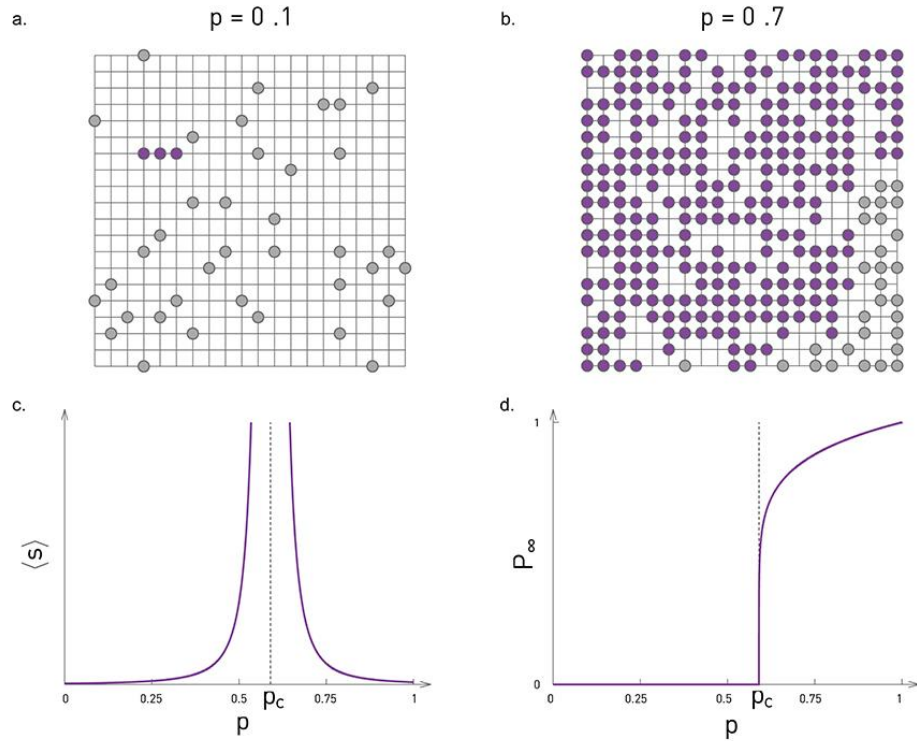
b) The larger p is, the larger the clusters are.

Cluster size does not change gradually with p . For a wide range of p , the lattice has many small clusters.

Barabasi (2016)

Can we measure network resilience/robustness?

When is a network a network?



Barabasi (2016)

Percolation theory

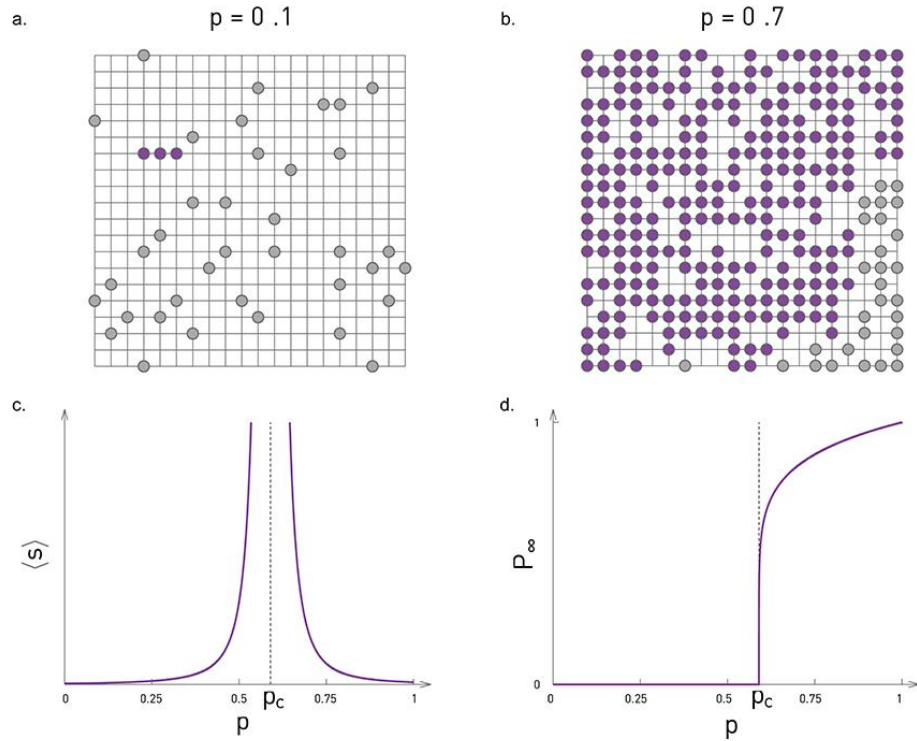
a) We place pebbles with probability p at each intersection. Neighbouring pebbles are considered connected, forming clusters of size two or more.

b) The larger p is, the larger the clusters are.

Cluster size does not change gradually with p . For a wide range of p , the lattice has many small clusters.

Can we measure network resilience/robustness?

When is a network a network?



Barabasi (2016)

Percolation theory

a) We place pebbles with probability p at each intersection. Neighbouring pebbles are considered connected, forming clusters of size two or more.

b) The larger p is, the larger the clusters are.

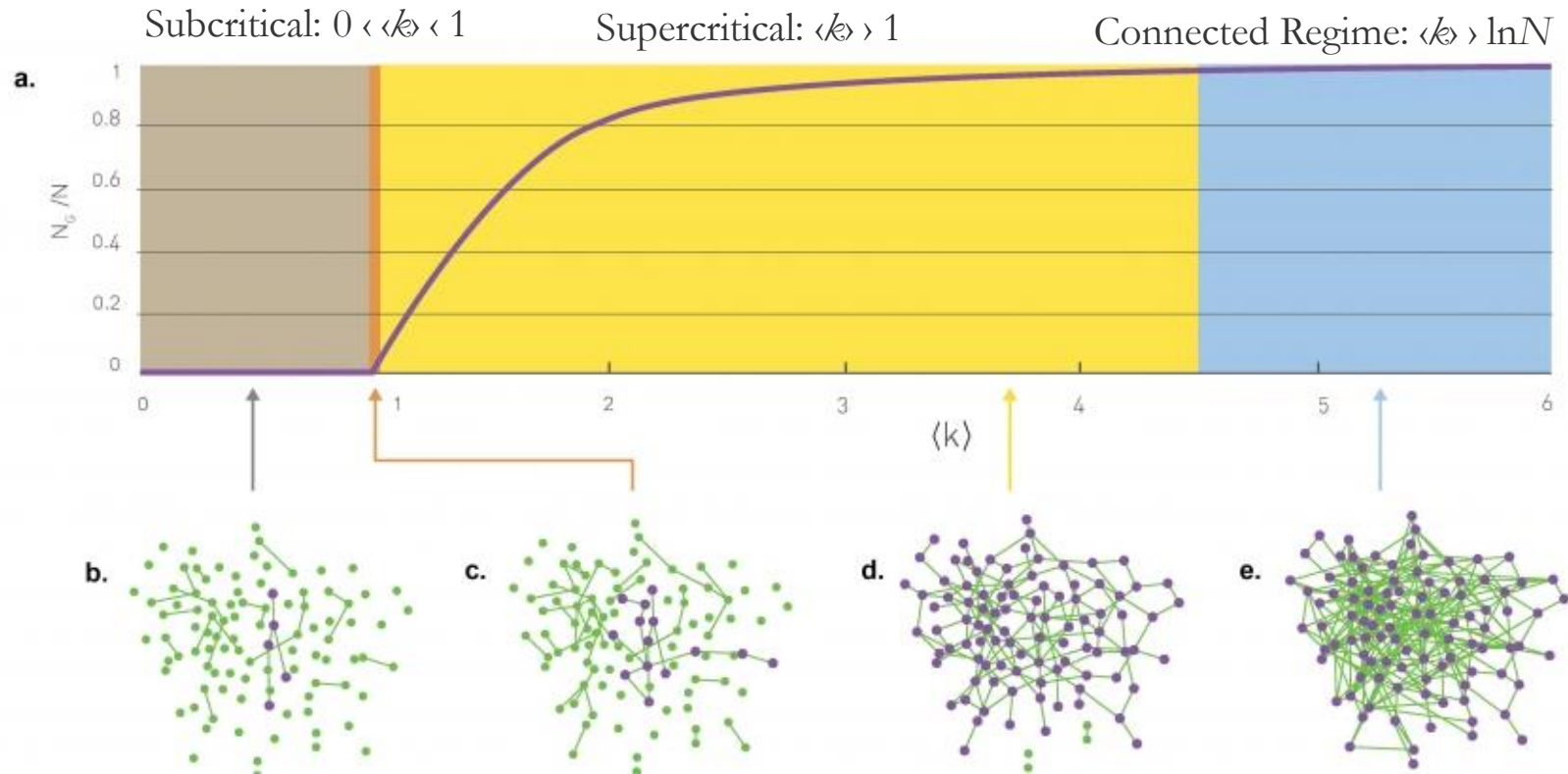
Cluster size does not change gradually with p . For a wide range of p , the lattice has many small clusters.

c) When p approaches a critical value p_c (specific to the lattice) from below, the disconnected clusters merge quickly.

d) The probability that a randomly chosen pebble belongs to the largest cluster (P_∞) takes positive values above the threshold p_c .

Can we measure network resilience/robustness?

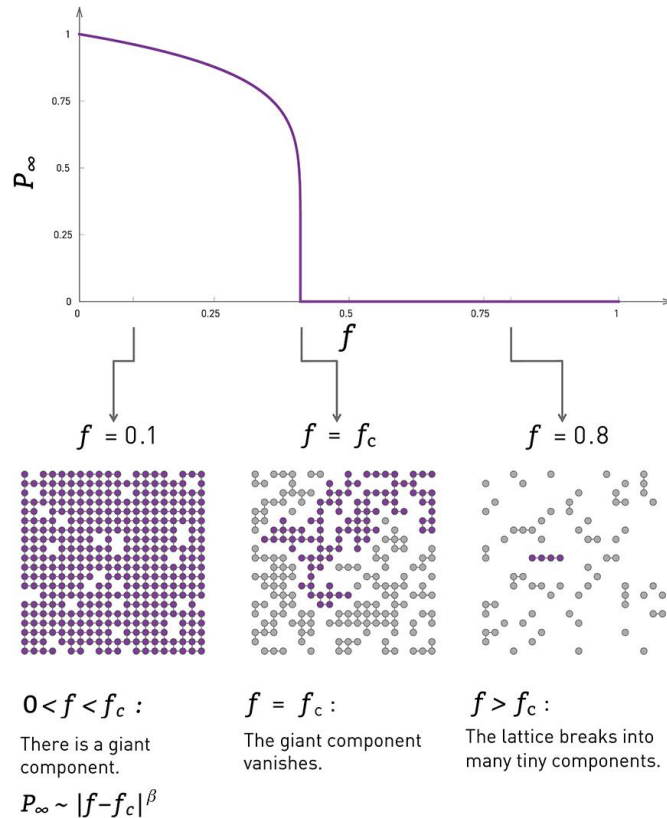
When is a network a network?



The relative size of the **giant component** in function of the average degree $\langle k \rangle$ in the Erdős-Rényi model.

Can we measure network resilience/robustness?

When is a network a network?



Barabasi (2016)

Inverse Percolation theory

Back to the main question: how many nodes do we have to remove to fragment a network into isolated components?

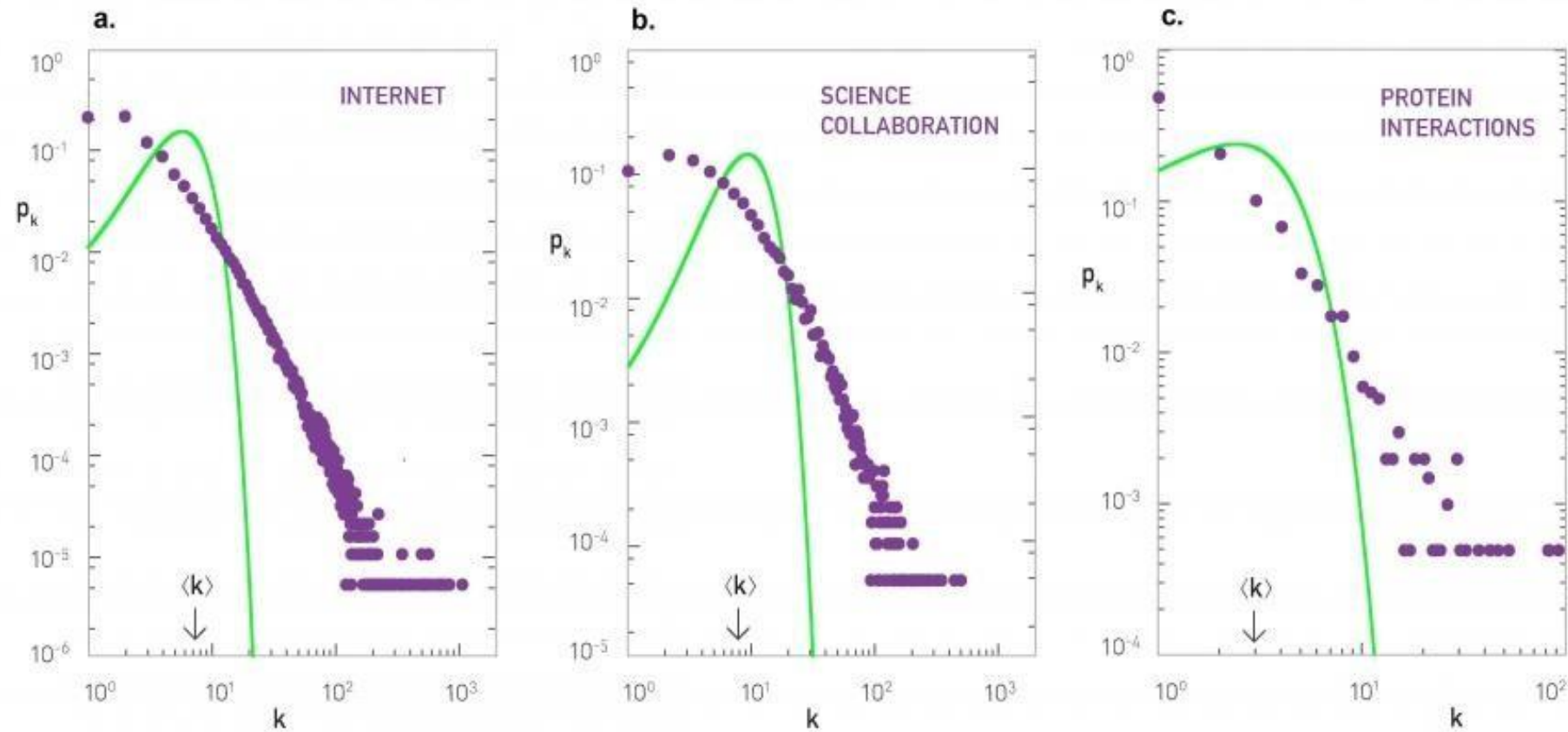
How does randomly removing f fraction of nodes impacts the integrity of the lattice (network)?

If the rate of node removal is below the critical threshold (f_c) the damage to the network's integrity is limited.

However, if f exceeds f_c , the **giant component** vanishes.

Can we measure network resilience/robustness?

Empirical networks are very different from ER networks! In particular they have heterogeneous degree distributions. How does this affect their robustness?

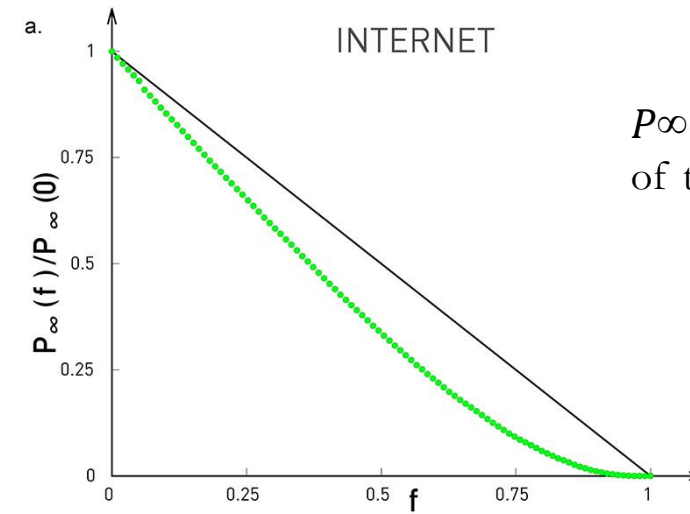


Can we measure network resilience/robustness?

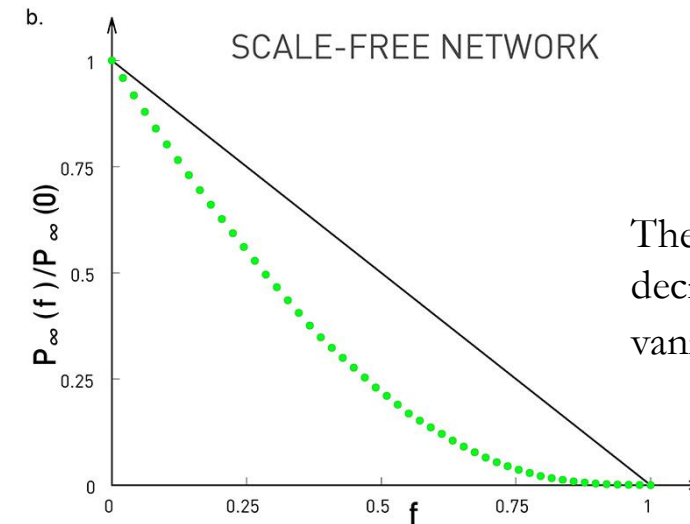
What happens when the network has a heterogeneous degree distribution?
How do the hubs affect robustness?

Example: the router-level Internet. Randomly select and remove nodes one-by-one. The Internet should fragment into many isolated subgraphs once the number of removed nodes reaches some critical value f_c .

The simulations indicate otherwise: the size of the largest component decreases gradually, vanishing only in the vicinity when we have removed practically all the nodes.



$P_{\infty}(f)/P_{\infty}(0)$ is the relative size of the giant component.



The largest component decreases in size gradually and vanishes only when f is near 1.

Can we measure network resilience/robustness?

In the case of the Erdos-Renyi network we know it is enough to remove so many nodes that the remaining network has average degree less than 1. It turns out there is a threshold for networks of arbitrary degree distribution to have a giant connected component:

$$\kappa = \frac{\langle k^2 \rangle}{\langle k \rangle} > 2$$

Idea: “For there to be a giant component, each node in that component must be connected to at least two other nodes”.

Idea: “For there to be a giant component, each node in that component must be connected to at least two other nodes”.

$$\kappa = \frac{\langle k^2 \rangle}{\langle k \rangle} > 2$$

In other word the expected degree of the node should be equal or exceed two, the condition for node i to be part of the giant component:

$$\langle k_i | i \leftrightarrow j \rangle = \sum_{k_i} k_i P(k_i | i \leftrightarrow j) = 2$$

We can rewrite the probability appearing in the sum using the properties of conditional probabilities:

Conditional probability of a node i of degree k is connected to a node j in the giant component.

$$P(k_i | i \leftrightarrow j) = \frac{P(k_i, i \leftrightarrow j)}{P(i \leftrightarrow j)} = \frac{P(i \leftrightarrow j | k_i) p(k_i)}{P(i \leftrightarrow j)}$$

$$P(i \leftrightarrow j | k_i) = \frac{k_i}{N-1}$$

←

↓

→

We can link to $N-1$ nodes, and we can try it k times.

$$P(i \leftrightarrow j) = \frac{2L}{N(N-1)} = \frac{\langle k \rangle}{N-1}$$

$$P(k_i | i \leftrightarrow j) = \frac{P(k_i, i \leftrightarrow j)}{P(i \leftrightarrow j)} = \frac{P(i \leftrightarrow j | k_i) p(k_i)}{P(i \leftrightarrow j)} = \sum_{k_i} k_i \frac{k_i p(k_i)}{\langle k \rangle}$$

With that we arrive at the Molloy-Reed criterion, providing the condition to have a giant component as: $\kappa = \frac{\langle k^2 \rangle}{\langle k \rangle} > 2$

Plugging in the second moment of the ER network’s degree distribution (Poisson): $\frac{\langle k^2 \rangle}{\langle k \rangle} = \frac{\langle k \rangle (1 + \langle k \rangle)}{\langle k \rangle} = 1 + \langle k \rangle > 2$

Plugging in the second moment of the ER network's degree distribution (Poisson): $\frac{\langle k^2 \rangle}{\langle k \rangle} = \frac{\langle k \rangle(1 + \langle k \rangle)}{\langle k \rangle} = 1 + \langle k \rangle > 2$



We recover that the average degree should be at least 1.

We can use this to define a critical threshold for when networks become disconnected under random failures:

$$f_c = 1 - \frac{1}{\frac{\langle k^2 \rangle}{\langle k \rangle} - 1}$$

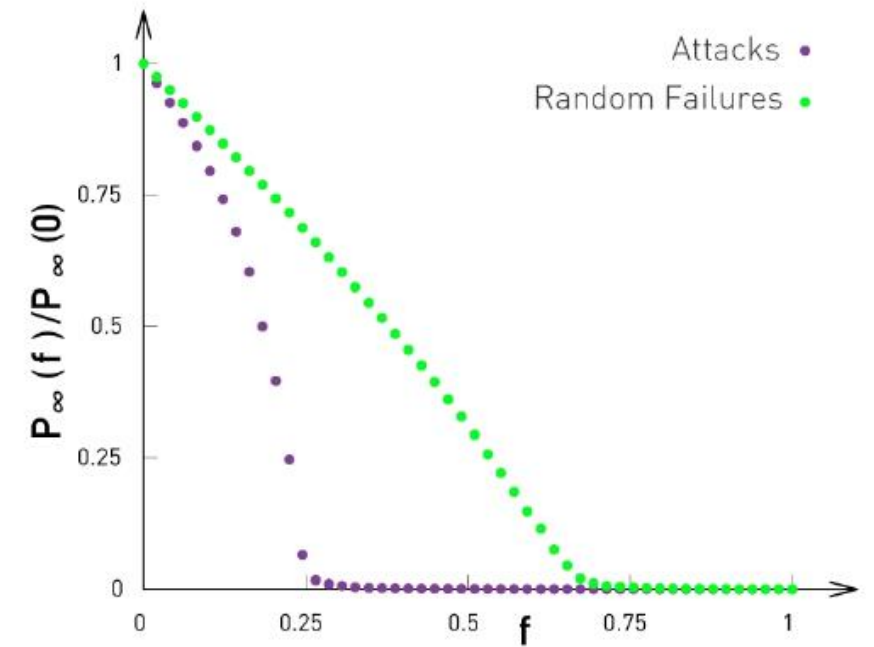
NETWORK	RANDOM FAILURES (REAL NETWORK)	RANDOM FAILURES (RANDOMIZED NETWORK)
Internet	0.92	0.84
WWW	0.88	0.85
Power Grid	0.61	0.63
Mobile-Phone Call	0.78	0.68
Email	0.92	0.69
Science Collaboration	0.92	0.88
Actor Network	0.98	0.99
Citation Network	0.96	0.95
E. Coli Metabolism	0.96	0.90
Yeast Protein Interactions	0.88	0.66

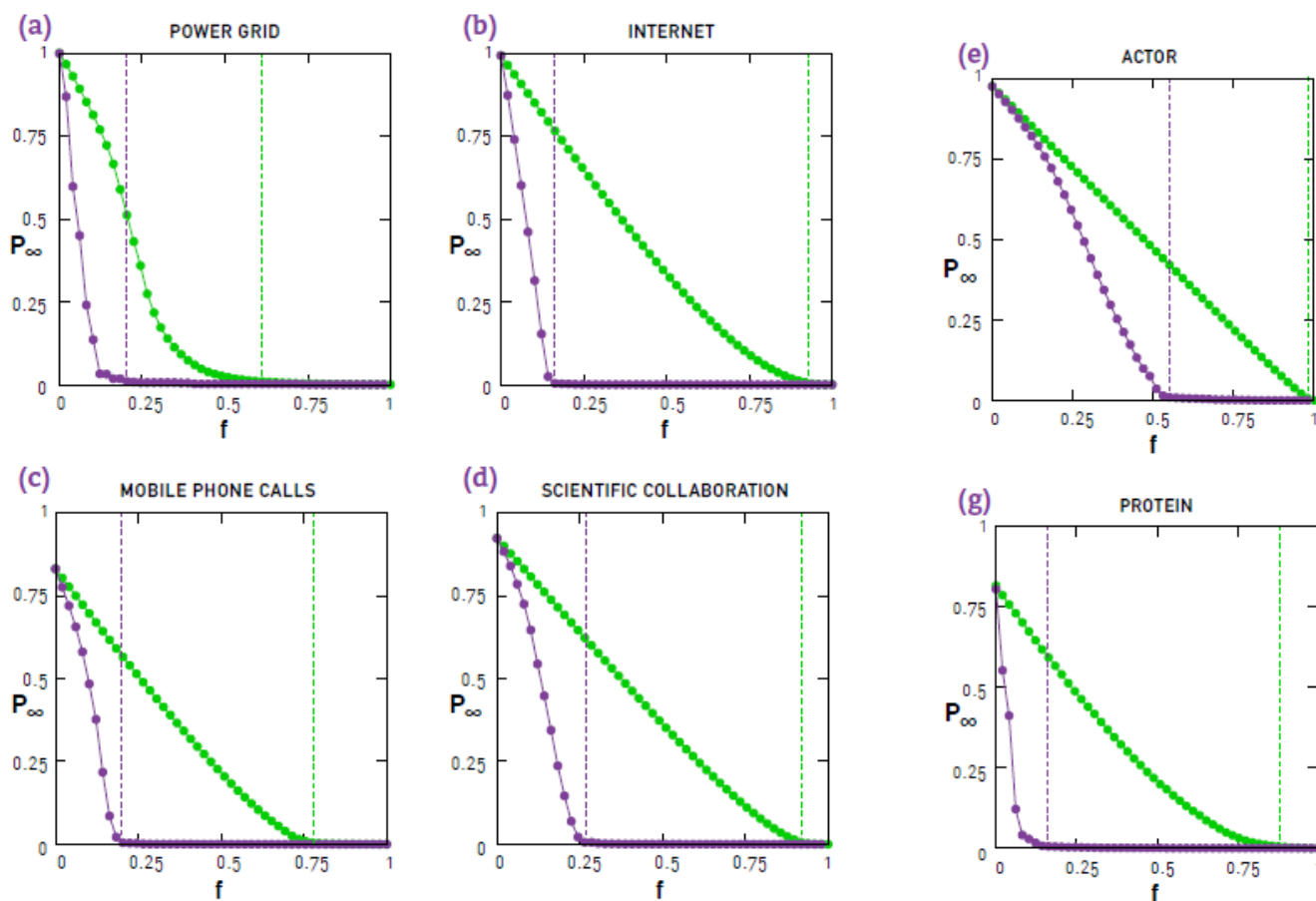
Can we measure network resilience/robustness **in case of non-random failures?**

What if, instead of removing nodes at random, we go after the hubs of a scale-free network?

Attacking the network this way assumes knowledge of its topology and the targeting of its hubs.

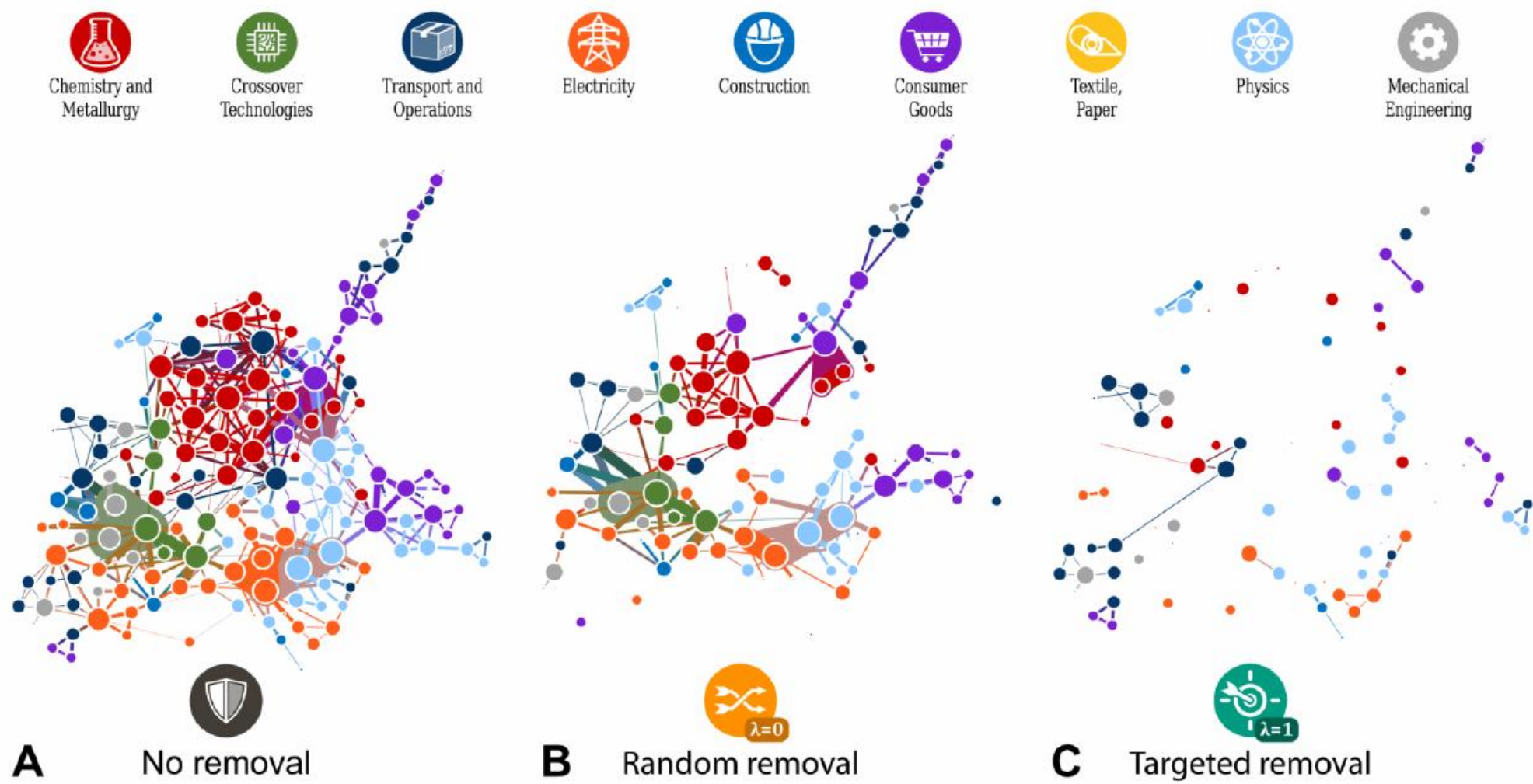
The critical threshold (f_c) emerges rapidly in this case.





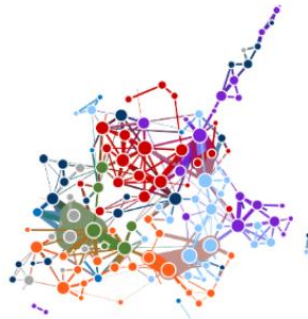
Network	Random Failures (Real Network)	Attack (Real Network)
Internet	0.92	0.16
WWW	0.88	0.12
Power Grid	0.61	0.20
Mobile Phone Calls	0.78	0.20
Email	0.92	0.04
Science Collaboration	0.92	0.27
Actor Network	0.98	0.55
Citation Network	0.96	0.76
E. Coli Metabolism	0.96	0.49
Protein Interactions	0.88	0.06

Predicting regions performance with resilience

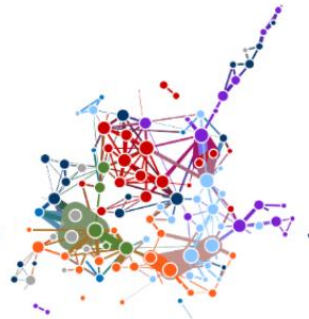




RANDOM REMOVAL



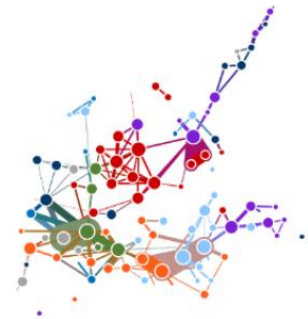
A) $\Omega^{\lambda=0} = 0.1 \frac{\langle k^0 \rangle}{\langle k^0 \rangle} = 0.898$



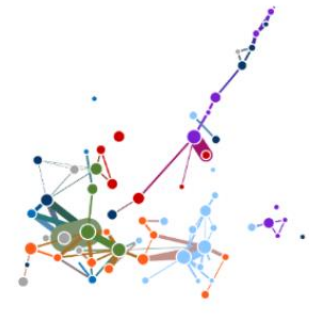
B) $\Omega^{\lambda=0} = 0.2 \frac{\langle k^0 \rangle}{\langle k^0 \rangle} = 0.801$



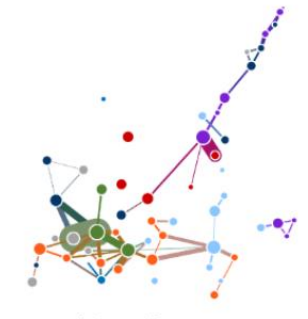
C) $\Omega^{\lambda=0} = 0.3 \frac{\langle k^0 \rangle}{\langle k^0 \rangle} = 0.703$



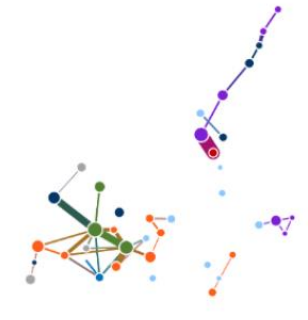
D) $\Omega^{\lambda=0} = 0.4 \frac{\langle k^0 \rangle}{\langle k^0 \rangle} = 0.593$



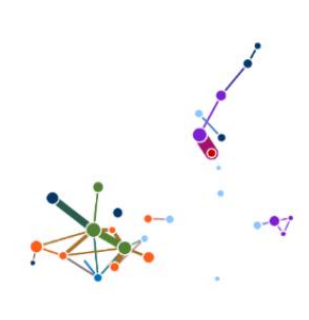
E) $\Omega^{\lambda=0} = 0.5 \frac{\langle k^0 \rangle}{\langle k^0 \rangle} = 0.512$



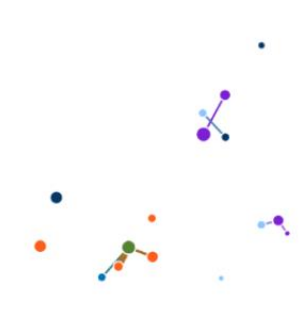
F) $\Omega^{\lambda=0} = 0.6 \frac{\langle k^0 \rangle}{\langle k^0 \rangle} = 0.388$



G) $\Omega^{\lambda=0} = 0.7 \frac{\langle k^0 \rangle}{\langle k^0 \rangle} = 0.306$



H) $\Omega^{\lambda=0} = 0.8 \frac{\langle k^0 \rangle}{\langle k^0 \rangle} = 0.205$



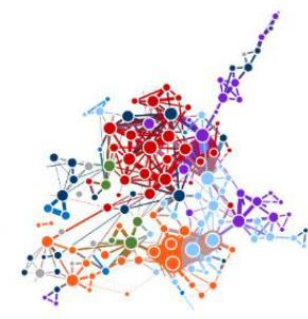
I) $\Omega^{\lambda=0} = 0.9 \frac{\langle k^0 \rangle}{\langle k^0 \rangle} = 0.078$



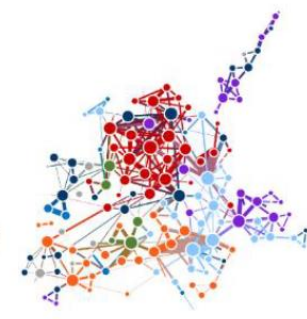
TARGETED REMOVAL



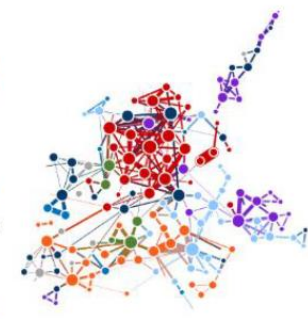
A) $\Omega^{\lambda=1} = 0.01 \frac{\langle k^0 \rangle}{\langle k^0 \rangle} = 0.909$



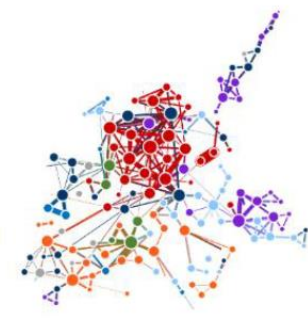
B) $\Omega^{\lambda=1} = 0.02 \frac{\langle k^0 \rangle}{\langle k^0 \rangle} = 0.841$



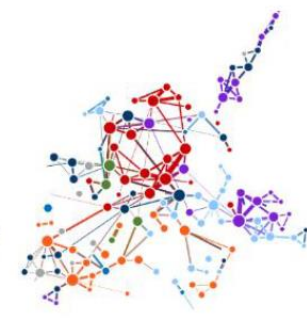
C) $\Omega^{\lambda=1} = 0.03 \frac{\langle k^0 \rangle}{\langle k^0 \rangle} = 0.778$



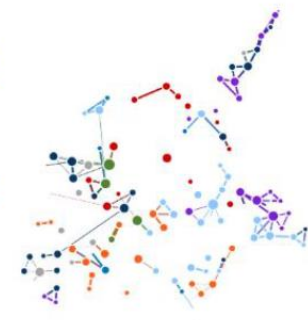
D) $\Omega^{\lambda=1} = 0.04 \frac{\langle k^0 \rangle}{\langle k^0 \rangle} = 0.720$



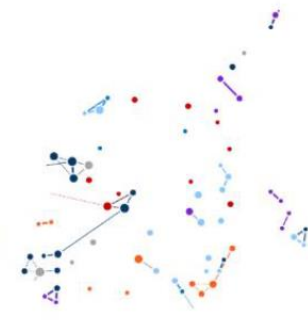
E) $\Omega^{\lambda=1} = 0.05 \frac{\langle k^0 \rangle}{\langle k^0 \rangle} = 0.673$



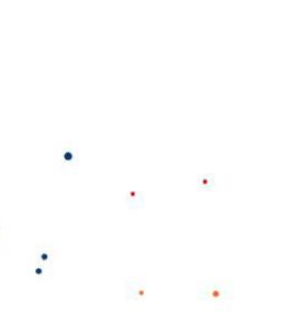
F) $\Omega^{\lambda=1} = 0.10 \frac{\langle k^0 \rangle}{\langle k^0 \rangle} = 0.501$



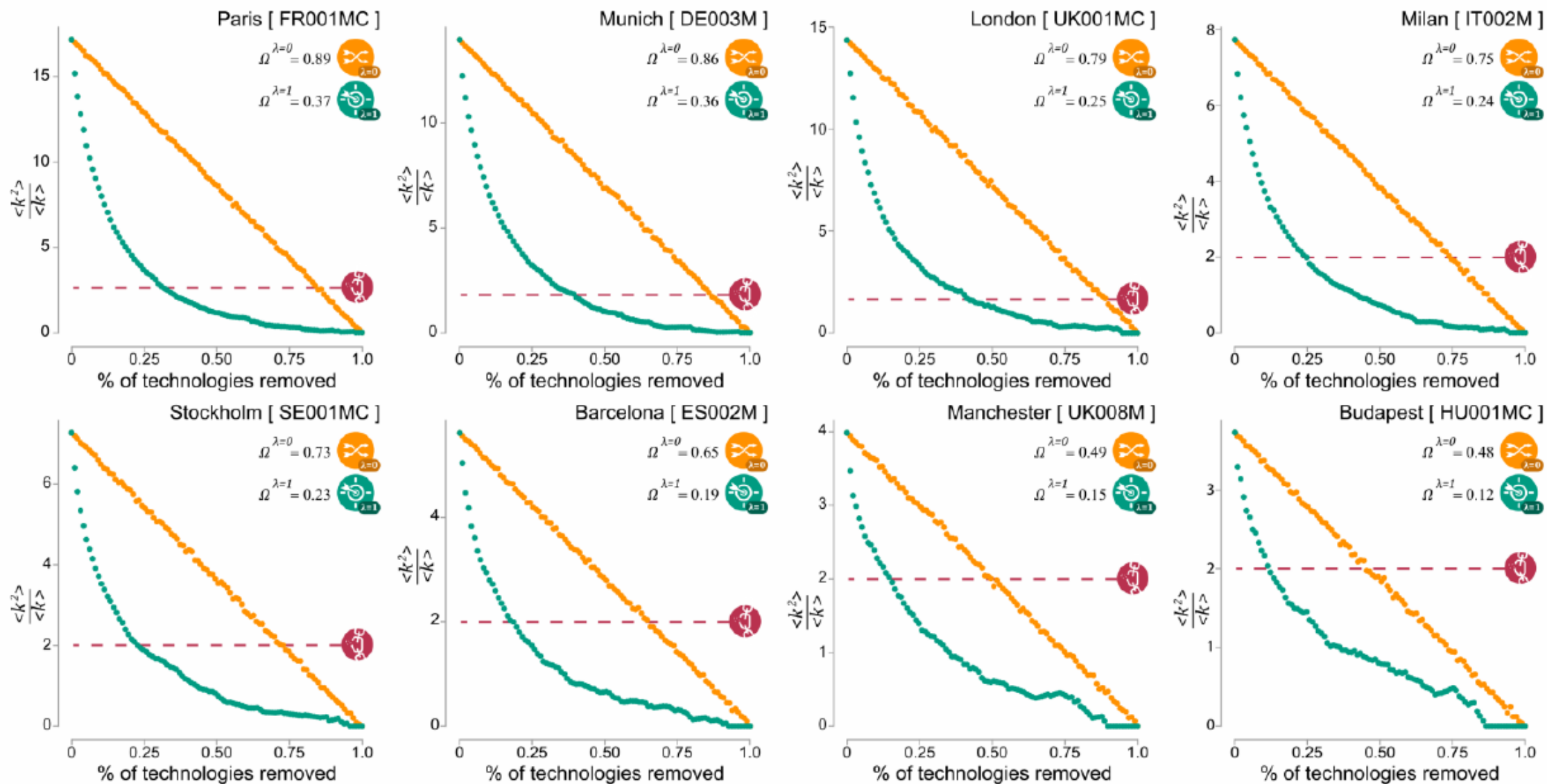
G) $\Omega^{\lambda=1} = 0.20 \frac{\langle k^0 \rangle}{\langle k^0 \rangle} = 0.316$

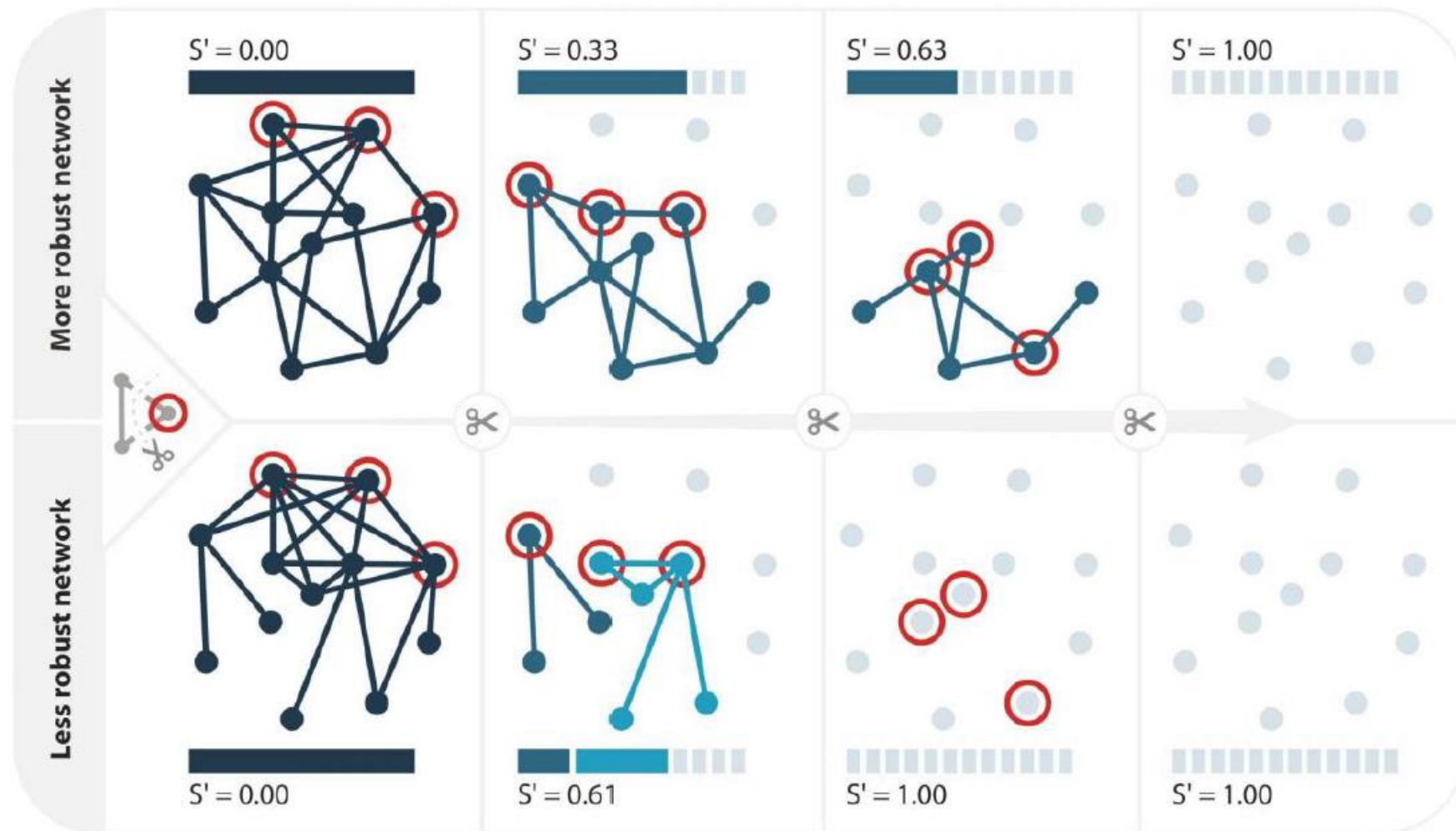


H) $\Omega^{\lambda=1} = 0.40 \frac{\langle k^0 \rangle}{\langle k^0 \rangle} = 0.176$



I) $\Omega^{\lambda=1} = 0.80 \frac{\langle k^0 \rangle}{\langle k^0 \rangle} = 0.082$





Calculate the Shannon entropy of node distribution across the isolated components:

$$S(G_i^f) = - \sum_{k=1}^K p_k \log p_k$$

Scale the Shannon entropy with the log number of industries present in the region:

$$S'(G_i^f) = S(G_i^f) / \log N$$

The final measure of robustness can be calculated as 1 minus the area under this curve:

$$\Omega(G_i) = 1 - \int_0^1 S'(G_i^f) df$$

500 runs of the LASSO variable selection:

$$\frac{\text{Emp}_{i,t+s}}{\text{Emp}_{i,t}} = \alpha + \text{Emp}_{i,t} + \gamma_1 \Omega(G_i)^{R/T} + \beta_1 [Z_{i,t}] + e_{i,t}$$

Files

main

+

Q

Go to file

> 00_course_introduction

> 01_graph_theory_and_network_i...

> 02_graph_theory_and_network_i...

> 03_graph_theory_and_network_i...

> 04_modelling_network_structure_I

> 05_modelling_network_structure_II

> 06_modelling_network_structure...

> 02_data

toy_elist_a.csv

toy_elist_b.csv

> 03_scripts

> assets

network_robustness.html

network_robustness.rmd

Albertetal(2000).pdf

Elekesetal(2024).pdf

Network_Homework_6.pdf

Network_Rajk2024_6.pdf

README.md

Schuessler-Wachs(2024).pdf

> 09_temporal_and_higher_order

> 10_diffusion_and_contagion

> 11_applied_network_science

rajk2024 / 06_modelling_network_structure_III / 03_scripts / network_robustness.rmd

CodeBlame583 lines (475 loc) · 17.9 KB

RawCopyDownloadEditDropdownFullscreen

```
104 The Molloy-Reed criterion (Molloy & Reed 1995) assumes that the network is non-assortative to degree.
105
106 * **Degree non-assortative network**: nodes link to each other randomly.
107 * **Degree assortative network**: higher degree nodes tend to link with other higher degree nodes.
108 * **Degree disassortative network**: higher degree nodes tend to link with lower degree nodes.
109
110 ```{r}
111 # Test degree correlation using Pearson's correlation
112 toy_elist_a_twoway <-
113   # Transform the edgelist to a two-way edgelist
114   toy_elist_a %>%
115   rename(
116     node_i = node_j,
117     node_j = node_i
118   ) %>%
119   bind_rows(toy_elist_a) %>%
120   distinct() %>%
121   # Get degrees
122   group_by(node_i) %>%
123     mutate(degree_i = n()) %>%
124   group_by(node_j) %>%
125     mutate(degree_j = n()) %>%
126   ungroup()
127
128 # Test
129 cor.test(
130   toy_elist_a_twoway$degree_i,
131   toy_elist_a_twoway$degree_j
132 )
133 ...
134
135 # Get network robustness
136
137 Below I present the measurement logic based on Elekes et al. (2024), which is in turn was based on Zitnik et al (2019).
138
139 * The  $i$ -th input network  $G_i = (V_i, E_i)$  correspond with a set of nodes  $V_i$  and a set of edges  $E_i$ .
140 *  $N$  is the number of nodes  $V_i$ ,  $M$  is the number of edges  $E_i$ .
141 *  $f \in [0,1]$  is the rate of the proportion of nodes removed from the network.
```

igraph & NetworkX

Thank you!

twitter: @Gergo_Toht

web: anetilab.org/researchers/g-toth

email: toth.gergo@krtk.elte.hu / gergo.toth@umu.se

	(1) All Sectors	(2) Industry	(3) All Sectors	(4) Industry	(5) All Sectors	(6) Industry
$\Omega^{\lambda=0}$			0.0594 (0.038)	0.1046*** (0.036)		
$\Omega^{\lambda=1}$					0.1618** (0.076)	0.2487*** (0.079)
$\log(GVA)$	-0.0569** (0.027)	-0.0656 (0.039)	-0.0504* (0.028)	-0.0607 (0.041)	-0.0469 (0.028)	-0.0562 (0.042)
$\log(POP)$	0.0159 (0.048)	-0.0139 (0.033)	0.0494 (0.056)	-0.019 (0.035)	0.0508 (0.055)	-0.0224 (0.035)
$\log(EMPRATE)$	0.0019 (0.038)	0.0065 (0.016)	-0.0371 (0.046)	0.0071 (0.017)	-0.0425 (0.043)	0.0053 (0.017)
Constant	1.2406*** (0.122)	1.4044*** (0.160)	1.2078*** (0.140)	1.4034*** (0.174)	1.1993*** (0.139)	1.3947*** (0.176)
Clustered SE	Yes	Yes	Yes	Yes	Yes	Yes
Mean VIF	3.51	3.51	3.38	3.38	3.12	3.12
R^2	0.192	0.165	0.209	0.191	0.216	0.195
Adj. R^2	0.173	0.146	0.184	0.166	0.192	0.170
Observations	269	269	269	269	269	269

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Reading and homework – Nov 25

Read Lengyel et al. Scientific Reports 2020

Homework

1. Identify a limitation/niche that this paper does not cover in the context of spatial innovation diffusion in social networks.
2. Prepare a **maximum 1-page** pdf document:
 - a) Describe this niche (1 paragraph),
 - b) Explain why covering this niche is important for science/society/business (1 paragraph),
 - c) Let me know if you have an idea for methods/data/context to address the niche (1 paragraph),
 - d) What could we learn from this new analysis (1 paragraph),
 - e) Add the references of within-text citations in case you do 2c and 2d as well.
3. Save the file: "givenname_familyname_EC2025_HW4.pdf". Send it to the instructor and TA by email. Deadline: December 1, Monday, 8am. Homeworks sent after the deadline are not considered.
4. Homeworks covering 2a-b can get 2.5 points. Including 2c-d-e can get you additional 1.5 points. Prepare for a discussion about your homework in the class to collect 1 „Activity point”.